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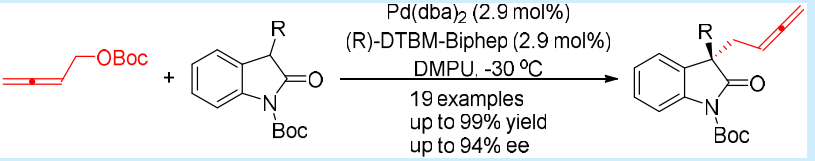
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Pd-Catalyzed Enantioselective Creation of All-Carbon Quaternary Center with 2,3-Allenylic Carbonates

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ABSTRACT: Enantioselective construction of all-carbon quaternary centers has been achieved via the palladium-catalyzed highly enantioselective allenylation of oxindoles with 2,3-allenylic carbonates to afford a variety of optically active allene products, which contain oxindole units with different functional groups, in high ee. The corresponding synthetic applications have also been demonstrated.

The wide occurrence of the allene unit in natural products, biologically active compounds, and marketed drugs,[1] as well as the diversified reactivities,[2] have promoted the rapid development of allene chemistry in the last three decades. Recently, much effort has been devoted to the asymmetric synthesis of allenes with a unique axial chirality.[3] One of the most straightforward methods would be the metal-catalyzed enantioselective allenylation of a nucleophile with racemic 2,3-allenol derivatives constructing an axial chirality and/or center chirality within the allene starting material (Scheme 1a, eq 1).[4−6] However, extensive studies on differently substituted 2,3-allenol derivatives have been focused on the construction of a chiral center and/or central chirality of the starting allenes, which are close to those of chiral ligands. Recently, there have been a few reports on enantioselective 2,3-butadienylation of chiral *N*-Cu-metalated azomethine ylides generating optically active 2,3-butadienyl α-amino acid derivatives using the synergistic Pd/Cu catalysis with either two different or the same chiral ligands (Scheme 1a, eq 2).[7] So far, there is no report on the enantioselective allenylation creating a chiral center on the nucleophiles by utilizing the chiral palladium catalyst alone (Scheme 1a, eq 3).[8] The challenge is that the to-be-formed stereocenter is relatively far from the chiral environment. On the other hand, oxindoles are prevalent in a wide array of natural products and bioactive compounds (Scheme 1c)[9] and are also important synthetic units in organic synthesis.[10] Thus, there is an urgent need to develop new protocols that provide new diversified oxindole compounds.[11] Here, we report the first enantioselective Pd-catalyzed allenylation reaction of oxindoles with 2,3-allenylic carbonates[12] providing optically active allene derivatives bearing an oxindole in which an all-carbon quaternary center was constructed in the 3-position of oxindoles (Scheme 1b).

Oxindole **1a** and 2,3-allenylic *tert*-butyl carbonate **2a** were selected as the model substrates. Different chiral bidentate bisphosphine ligand skeletons were screened with $Pd(dba)_2$ as the palladium source for the reaction in *N*,*N*′-dimethylpropyleneurea (DMPU) at −30 °C [Table 1, (*R*)-**L1**-(*R*)-**L7**]. (*R*)-**L7** was found to be the best ligand and provided the product (*R*)-**3aa** in 50% yield with −39% ee. Then the P-phenyl group of (*R*)-**L7** was replaced with other substituted phenyl groups [Table 1, (*R*)-**L8**-(*R*)-**L12**], and it was found that (*R*)-**L12** with 3,5-(*t*-Bu)$_2$-4-MeO-phenyl is the best by providing the product (*R*)-**3aa** in 92% yield and 91% ee.

After further optimization, it was observed that when 2.9 mol % $Pd(dba)_2$ and 2.9 mol % (*R*)-DTBM-Biphep [(*R*)-**L12**] were applied, the reaction in DMPU at −30 °C afforded the coupling product (*R*)-**3aa** in the highest yield and enantioselectivity. The scope of oxindoles was then investigated with 2,3-allenylic *tert*-butyl carbonate **2a** (Table 2). The R^1 can be aryl groups substituted with 2-methyl, 2-ethyl, 2-methoxy, 2-isopropyl, 2,4-dimethyl, 2,5-dimethyl, 4-methoxy-2-methyl, and 4-chloro-2-methyl groups to afford the products (*R*)-**3aa**-(*R*)-**3ha** in good yields and high enantioselectivity (87%−97% yields, 58%−92% ee, Table 2, entries 1−8). The absolute configuration of the product was determined by the single-crystal X-ray diffraction study of (*R*)-**3da** (Figure S2). Various substituents (R^2) on the aromatic rings were well tolerated. Electron-donating groups such as *p*-Me- and *p*-OMe-containing substrates afforded the

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Scheme 1. Enantioselective Allenylation of Nucleophiles and Pd-Catalyzed Reactions of 2,3-Allenol Derivatives with Nucleophiles

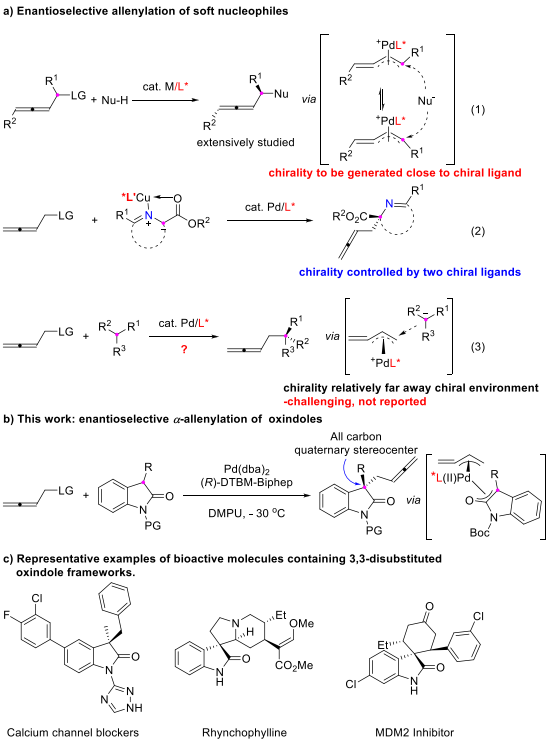
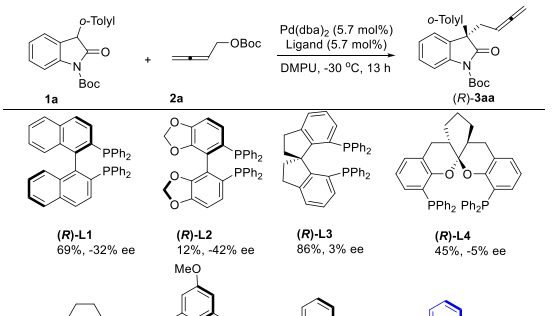


Table 1. Optimization of Pd-Catalyzed Enantioselective Allenylation Reaction of Oxindole 1a with 2a[a]



[a]Reaction conditions: The reaction of **1a** (0.088 mmol), **2a** (0.1 mmol), $Pd(dba)_2$ (5.7 mol %), and ligand (5.7 mol %) in DMPU (0.5 mL) was conducted at −30 °C. NMR yield determined with CH_2Br_2 as the internal standard. The ee was determined by chiral HPLC analysis after column chromatographic separation on silica gel. [b]The reaction time was 12 h.

corresponding products (*R*)-**3ia** and (*R*)-**3ja** in 97% and 69% yields, respectively, with an enantiomeric excess of 93% (Table 2, entries 9 and 10). The benzene ring of 3-*o*-tolyl-oxindoles

Table 2. Substrate Scope of the Pd-Catalyzed Enantioselective Allenylation of Oxindoles[a]



			(*R*)-3, %	
entry	R^1	R^2	yield[b]	ee[b]
1	2-MeC_6H_4 (**1a**)	H	92% [(*R*)-**3aa**]	91%
2	2-EtC_6H_4 (**1b**)	H	87% [(*R*)-**3ba**]	91%
3	2-$MeOC_6H_4$ (**1c**)	H	94% [(*R*)-**3ca**]	58%
4	2-*i*PrC_6H_4 (**1d**)	H	95% [(*R*)-**3da**]	85%
5	2-Me-4-MeC_6H_3 (**1e**)	H	97% [(*R*)-**3ea**]	92%
6	2-Me-5-MeC_6H_3 (**1f**)	H	93% [(*R*)-**3fa**]	92%
7	4-MeO-2-MeC_6H_3 (**1g**)	H	92% [(*R*)-**3ga**]	92%
8	4-Cl-2-MeC_6H_3 (**1h**)	H	93% [(*R*)-**3ha**]	87%
9	2-MeC_6H_4 (**1i**)	5-Me	97% [(*R*)-**3ia**]	93%
10	2-MeC_6H_4 (**1j**)	5-MeO	69% [(*R*)-**3ja**]	93%
11	2-MeC_6H_4 (**1k**)	5-F	90% [(*R*)-**3ka**]	94%
12	2-MeC_6H_4 (**1l**)	5-Cl	91% [(*R*)-**3la**]	94%
13	2-MeC_6H_4 (**1m**)	5-Br	90% [(*R*)-**3ma**]	93%
14[c]	2-MeC_6H_4 (**1n**)	6-Cl	92% [(*R*)-**3na**]	88%
15[d]	2-MeC_6H_4 (**1n**)	6-Cl	90% [(*R*)-**3na**]	89%
16	2-MeC_6H_4 (**1o**)	6-Br	87% [(*R*)-**3oa**]	91%
17[e]	2-MeC_6H_4 (**1o**)	6-Br	99% [(*R*)-**3oa**]	92%
18[c]	2-MeC_6H_4 (**1p**)	7-Cl	88% [(*R*)-**3pa**]	87%
19[d,f]	2-MeC_6H_4 (**1p**)	7-Cl	85% [(*R*)-**3pa**]	91%
20	2-MeC_6H_4 (**1q**)	7-CF_3	86% [(*R*)-**3qa**]	83%
21[g]	*i*-Pr (**1r**)	H	73% [(*R*)-**3ra**]	63%
22[h]	Cy (**1s**)	H	58% [(*R*)-**3sa**]	64%

[a]Reaction conditions: The reaction of **1** (1.0 mmol), **2a** (1.14 mmol), $Pd(dba)_2$ (2.9 mol %), and (*R*)-DTBM-Biphep (2.9 mol %) in DMPU (5.0 mL) was conducted at −30 °C. [b]Isolated yield. The ee was determined by chiral HPLC analysis after column chromatographic separation on silica gel. [c]The reaction of **1** (0.088 mmol), **2a** (0.1 mmol), $Pd(dba)_2$ (5.7 mol %), and (*R*)-DTBM-Biphep (5.7 mol %) in DMPU (0.5 mL) was conducted at −30 °C. NMR yield determined by ^1H NMR analysis using CH_2Br_2 as the internal standard. [d](*R*)-DTBM-Segphos (2.9 mol %) was used instead of (*R*)-DTBM-Biphep (2.9 mol %). [e]The reaction of **1o** (5.0 mmol), **2a** (5.7 mmol), $Pd(dba)_2$ (2.9 mol %), and (*R*)-DTBM-Biphep (2.9 mol %) in DMPU (25.0 mL) was conducted at −30 °C. [f]The reaction was conducted at −28 °C. [g]DMPU (10.0 mL). [h]The reaction of **1s** (1.2 mmol), **2a** (1.0 mmol), $Pd(dba)_2$ (5.7 mol %), and (*R*)-DTBM-Biphep (5.7 mol %) in DMPU (5.0 mL) was conducted at −30 °C.

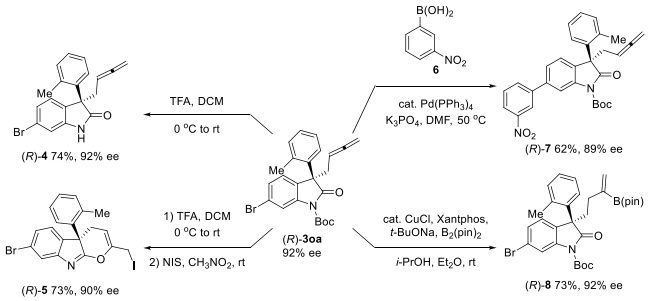
could also be substituted with 5-halo, 6-halo, 7-chloro, and 7-CF_3 groups, which provided the desired products (*R*)-**3ka**-(*R*)-**3qa** in yields of 85%−99% and 83%−94% ee (Table 2, entries 11−20). The reaction of oxindole **1o** with 2,3-allenylic *tert*-butyl carbonate **2a** can be easily executed on a 5 mmol scale to provide the final product (*R*)-**3oa** in 2.2505 g (99% yield and 92% ee, Table 2, entry 17). However, the enantioselectivity for 3-alkyl-substituted oxindoles (*R*)-**3ra** and (*R*)-**3sa** were moderate (63% and 64% ee) (Table 2, entries 21 and 22).

As a class of highly useful compounds in organic synthesis, the final product (*R*)-**3oa** can be easily transformed to other multifunctional chemicals with high ee values (Scheme 2): the *N*-substituent, *tert*-butyl carbonate, can be removed to provide the compound (*R*)-**4** without racemization.[13] The sequential treatment of (*R*)-**3oa** with TFA and *N*-iodosuccinimide (NIS) afforded the tricyclic iodide (*R*)-**5** in 73% yield and 90% ee. The Suzuki coupling reaction of (*R*)-**3oa** with 3-nitrophenyl boronic

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Scheme 2. Synthetic Transformations



acid **6** provided the coupling product (*R*)-**7**.[14] Finally, the borylcupration of the terminal allene unit in (*R*)-**3oa** with $B_2(pin)_2$ provided the alkenyl boronate product (*R*)-**8**.[15]

In conclusion, we have described a general method of the Pd-catalyzed enantioselective allenylation reaction of oxindoles with 2,3-allenylic carbonates for the efficient construction of optically active allene derivatives bearing an oxindole unit. The reaction works with a wide range of substrates to give the products in good yield with high enantioselectivity. The synthetic potentials of the final products have also been demonstrated. Further studies are being pursued in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.2c02497.

Experimental procedures, characterization for obtained compounds, crystal data for (*R*)-**3da** and (*R*)-**5**, NMR spectra, and HPLC charts (PDF)

Accession Codes

CCDC 1958258 and 1970013 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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